

MultiTaskMRINet: A LCNNet-050-Based Dual-Task Framework for Joint Brain MRI Authenticity Verification and Tumor Classification with Adversarial Robustness

Muhammad Rehan Tariq
Dept. of Artificial Intelligence
FAST-NUCES, Islamabad
muhammedrehantariq786@gmail.com

Abuzar Mehdi
Dept. of Artificial Intelligence
FAST-NUCES, Islamabad
abuzarmehdi214@gmail.com

Abstract—Due to the proliferation of generative models, in particular Generative Adversarial Networks (GANs) and adversarial attack algorithms such as PGD and FGSM, a critical security weakness has emerged in clinical neuroimaging: the capability to synthesise photorealistic brain MRI scans that are indistinguishable from real clinical studies. Current detection systems take care of this menace singly, with the execution of binary authenticity classification not accompanied by tumour-type classification. We introduce *MultiTaskMRINet*, a re-implemented slim multi-task convolutional neural network that uses an LCNNet-050 backbone. The new structure is a collaborative effort: two clinically relevant tasks may be jointly addressed in a single forward pass: (1) *authenticity verification* (real vs. fake, covering both GAN-generated and adversarially disturbed inputs), and (2) *tumour classification* (four-class: no-tumour, glioma, meningioma, pituitary). The shared backbone produces an embedding of 512 dimensions with the help of a Batch Normalisation-ReLU-Dropout block that feeds two independent classification heads, each with 256-dimensional hidden layers. A three-stage training curriculum backbone-frozen authenticity warm-up, tumour-head specialisation, and complete joint fine-tuning gradually opens up model capacity with pretrained feature richness preserved. Label smoothing, MixUp augmentation, and cosine annealing with AdamW further enhance generalisation. Grad-CAM saliency maps are computed on the final convolutional block of the authenticity head, offering radiologist-readable spatial explanations of every authenticity decision.

Index Terms—Medical deepfake detection, adversarial attack detection, multi-task learning, LCNNet-050, brain MRI, tumour classification, explainable AI, Grad-CAM, staged training, FGSM, PGD

I. INTRODUCTION

A basic component of neurological diagnosis is brain MRI, on which the foundation of clinical judgments is laid from surgery planning to radiotherapy targeting. The wholeness of MRI information is, as a result, a requirement to achieve safe clinical decision-making. Unfortunately, current generative models especially GANs [4] and denoising diffusion probabilistic models [7] can now engender tumour-bearing MRI images of sufficient fidelity to fool trained radiologists.

Ali *et al.* [1] demonstrated that adversarially caused additions and removals of tumours in brain MRI scans are clinically plausible and pose a real-life risk to patient safety, insurance integrity, and clinical-trial data provenance.

Along with the threats which are founded on synthesis, there are gradient-based adversarial attacks [5], [6] Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) which inject imperceptible pixel-level perturbations that cause neural classifiers to fail silently. Such attacks can transform, when applied to MRI data, a scan of a real tumour into one that is classified by a deployed detector as normal, or the contrary. This type of threat is mostly non-existent in previously known literature on detection.

There are two types of generative attack which are the most hazardous. During a *tumour insertion attack*, the scan of a healthy patient is altered in order to present a simulated lesion, which leads to unnecessary surgeries. In a *tumour removal attack*, a real pathological finding is erased, causing life-threatening diagnostic delays. Both of them have been proved with the help of publicly available GAN architectures [2], [3].

Prior detection work is extremely limited in scope. Ali *et al.* [1] achieved 91.5% binary accuracy with a ResNet50+HOG+SVM pipeline, but with no spatial localisation, no GAN-source label, and no explainability output. Grabovski *et al.* [2] proposed an unsupervised diffusion-based anomaly detector achieving AUC 0.90 (tumour injection) and 0.96 (tumour removal), but it cannot label the type of GAN and requires expensive iterative diffusion inference. The systematic review by Manzoor *et al.* [3] singles out XAI integration, spatial localisation, and GAN-source classification long with computationally expensive architectures which are not suitable for clinical deployment, as four prevailing open issues in the area. None of the sources reviewed are committed to adversarial-attack detection together with tumour classification.

Our work directly contributes the following:

- 1) A **re-architected lightweight multi-task architecture** (MultiTaskMRINet) with **LCNet-050** as the backbone, collaboratively solving authenticity detection (including GAN fakes, PGD attacks, and FGSM attacks) and tumour classification in a single forward pass.
- 2) A **larger fake dataset** which explicitly incorporates adversarial perturbations (PGD, FGSM) and GAN-generated samples, resulting in an 80/20 real-to-fake ratio throughout approximately 10,000–12,000 MRI slices.
- 3) A **three-stage training curriculum** that gradually liberates the backbone, allowing convergent stability in both tasks without catastrophic forgetting.
- 4) An **XAI module** based on Grad-CAM with spatial saliency maps consistent with manipulation areas, enabling radiologist-interpretable explanations of authenticity decisions.
- 5) An **end-to-end training pipeline** that includes label smoothing cross-entropy, MixUp augmentation, AdamW with cosine annealing, and automatic mixed-precision (AMP) training.

The rest of this paper is structured in the following way. Section II has related work reviewed. Section III outlines the proposed architecture. Section IV describes the experimental setup. Results are given in Section V. Section VI discusses findings and limitations. Section VII concludes.

II. BACKGROUND AND RELATED WORK

A. Medical Deepfake Generation

Generative Adversarial Networks were introduced by Goodfellow *et al.* [4] and quickly became adapted to medical imaging. DCGAN [16] and StyleGAN2 [17] generate high-fidelity 2-D image synthesis. CycleGAN [18] enables unpaired image-to-image translation allowing lesion insertion without matched scan pairs. More recently, denoising diffusion probabilistic models [7] have exceeded the quality of GAN on several benchmarks and are increasingly applied to medical image synthesis [14], [15]. Pradeepan *et al.* [12] and Sathishkumar *et al.* [13] provide detailed taxonomies of both creation and detection methods in the medical field.

B. Adversarial Attacks on Medical Imaging

Gradient-based adversarial attacks constitute a different and little-known threat in medical imaging. The Fast Gradient Sign Method [5] forms perturbations in a single gradient step, and Projected Gradient Descent [6] iteratively minimises model loss with an ℓ_∞ constraint. Unlike GAN-created fakes, adversarial examples need not have a generative model; they are directly calculated from the victim classifier, which causes them to be adaptable to deployed systems. Our work is the first, as far as we know, to include PGD- and FGSM-perturbed brain MRI scans as explicit fake classes in a multi-task detection system.

C. Detection Approaches

One can classify detection methods into three families. *Frequency-domain* techniques examine DCT/DFT artefacts introduced by up-sampling layers of GAN decoders. *CNN-based*

methods learn spatial discriminative features end-to-end; the dominating variants are ResNet [9] and DenseNet [10] variants in the literature [3]. *Diffusion-based anomaly detection* [2] re-uses a trained diffusion model as an anomaly scorer by measuring the reconstruction error of the reversing process.

Ali *et al.* [1] reflect the present state-of-the-art in supervised binary detection: they present an AFFETDS system that combines ResNet50 detailed features with Histogram of Oriented Gradients (HOG) descriptors, which classify using a weighted ensemble SVM. AFFETDS is however constrained by a small training set (1,378 samples), no spatial localisation, and incompatibility with distributed-scale implementation.

D. Explainability in Medical AI

Gradient-weighted Class Activation Mapping (Grad-CAM) was proposed by Selvaraju *et al.* [11], offering class-discriminative localisations of deep networks without architectural modification. Grabovski *et al.* [2] utilise XAI tools post-hoc but do not rely on saliency maps as a training signal. Our work integrates Grad-CAM directly into the inference pipeline, giving an explanation of each authenticity decision without additional overhead.

E. Efficient CNN Architectures

LCNet (Lightweight Classification Network) [8] is a lightweight backbone which uses depthwise separable convolutions and Hard-Swish activations optimised for low-FLOP inference on edge hardware. LCNet-050 uses a $0.5\times$ channel multiplier, which produces a parameter count under 2 M while maintaining competitive ImageNet accuracy. Its regularised feature hierarchy is well adapted to the distribution difference between natural images and grayscale MRI slices, making it a strong choice for medical imaging tasks.

III. PROPOSED MODEL: MULTITASKMRINET

A. Overview

MultiTaskMRINet is built around a single guiding principle: a common representation of low-level features should at the same time fulfil a variety of clinically significant tasks. The refined architecture contains four modules: (i) an LCNet-050 feature extractor, (ii) a common embedding layer with Batch Normalisation, (iii) two independent task-specific classification heads, and (iv) a Grad-CAM XAI module.

The major changes compared to v1:

- **Backbone:** Fine tuned LCNet-050 (through `timm`), with output dimensionality estimated dynamically at construction time, ensuring compatibility with any `timm` backbone.
- **Task reduction:** The three-head design (authenticity, tumour, fake-type) is rationalised to two heads (authenticity, tumour). Adversarial attacks (PGD, FGSM) and fake images produced by GANs are combined into one binary authenticity signal, easing the downstream clinical decision boundary.
- **Shared embedding:** Increased from 256-d to 512-d with explicit Batch Normalisation for better training stability.

- **Head architecture:** Each head now applies FC(emb $\rightarrow$ 256)–BatchNorm–ReLU–Dropout(0.5)–FC(256 $\rightarrow$ C).
- **Training:** A three-stage curriculum with backbone freezing, label smoothing, and MixUp augmentation, absent from v1, is introduced.

The entire architecture is shown in Figure 1.

B. Input Representation

Each input is a 2-D axial brain MRI slice of size 224×224 pixels. Per-image instance normalisation is performed on the raw images:

$$I_{\text{norm}} = \frac{I - \mu_I}{\sigma_I + \epsilon} \cdot 50 + 128, \quad \epsilon = 10^{-6},$$

clipped to $[0, 255]$ to eliminate inter-scanner intensity drift before the standard ImageNet normalisation ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Greyscale slices are repeated to three channels to allow ImageNet-pretrained weights.

Online augmentation during training includes horizontal flips ($p = 0.5$), vertical flips ($p = 0.2$), rotation up to $\pm 20^\circ$ ($p = 0.7$), shift-scale-rotate ($p = 0.4$), elastic transform ($\alpha = 120$, $\sigma = 6$, $p = 0.3$), brightness and contrast jitter (± 0.2 , $p = 0.6$), Gaussian noise ($p = 0.4$), Gaussian blur ($p = 0.2$), CLAHE ($p = 0.3$), random gamma ($p = 0.3$), and CoarseDropout (8 holes, 16×16 , $p = 0.2$).

C. Backbone: LCNet-050

The backbone is LCNet-050 pretrained on ImageNet-1k, loaded using `timm` with the classifier head removed and global average pooling enabled. LCNet-050 follows an inverted-residual bottleneck pattern with depthwise separable convolutions and Hard-Swish activations. Its $0.5\times$ channel multiplier produces a compact feature extractor with ≈ 1.9 M parameters and ≈ 46 M FLOPs for a 224×224 input. The backbone output dimension is deduced during model construction through a dummy forward pass, making the architecture backbone-agnostic:

$$d_{\text{bb}} = \text{dim}(\text{LCNet-050}(\mathbf{0}_{1 \times 3 \times 224 \times 224})). \quad (1)$$

Figure 2 illustrates the complete LCNet-050 feature extraction pipeline, from raw MRI input through the four depthwise bottleneck stages to the shared embedding fed into the task heads.

D. Shared Embedding

The backbone output $\mathbf{f} \in \mathbb{R}^{d_{\text{bb}}}$ is mapped to a fixed 512-d shared embedding:

$$\mathbf{z} = \text{Dropout}_{0.5}(\text{ReLU}(\text{BN}(\text{FC}_{d_{\text{bb}} \rightarrow 512}(\mathbf{f})))), \quad \mathbf{z} \in \mathbb{R}^{512}. \quad (2)$$

Batch Normalisation before the non-linearity offers gradient-flow stability and makes it less sensitive to the choice of learning rate; an important advantage in the staged training setting detailed in Section IV.

E. Multi-Task Classification Heads

Two independent heads branch from $\mathbf{z}$. Each head follows the structure: FC(512 $\rightarrow$ 256)–BatchNorm–ReLU–Dropout(0.5)–FC(256 $\rightarrow$ C).

1) *Authenticity Head* ($C = 2$):

$$\hat{y}_{\text{auth}} = \text{Softmax}(W_A \mathbf{h}_A + b_A), \quad W_A \in \mathbb{R}^{2 \times 256},$$

where $\mathbf{h}_A = \text{ReLU}(\text{BN}(\text{FC}(\mathbf{z})))$, classifying inputs as *Real* (0) or *Fake* (1). *Fake* includes GAN-generated samples, PGD-perturbed scans, and FGSM-perturbed scans.

2) *Tumour Classification Head* ($C = 4$):

$$\hat{y}_{\text{tumor}} = \text{Softmax}(W_T \mathbf{h}_T + b_T), \quad W_T \in \mathbb{R}^{4 \times 256},$$

classifying one of four classes: *No Tumour*, *Glioma*, *Meningioma*, and *Pituitary*. This head is supervised only on samples labelled *Real*; its loss is masked to zero for fake inputs.

Figure 3 illustrates the internal structure of both heads and the conditional routing logic.

F. Conditional Inference Routing

During test time, the authenticity head is tested first. If the predicted label is *Real* ($\hat{y}_{\text{auth}} = 0$), inference proceeds through the tumour head to form a clinical tumour label. When the predicted value is *Fake* ($\hat{y}_{\text{auth}} = 1$), the tumour head is suppressed, and the system returns only the binary fake label.

The joint label space is:

$$\mathcal{Y}_{\text{joint}} = \{\text{no-tumour, glioma, meningioma, pituitary, fake}\},$$

a five-class space as opposed to the six-class space in v1.

G. Joint Training Objective

The two heads are jointly optimised with Label Smoothing Cross-Entropy (LSCE) using smoothing factor $\epsilon_s = 0.1$:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{auth}}^{\text{LSCE}} + \beta \cdot \mathcal{L}_{\text{tumor}}^{\text{LSCE}}, \quad (3)$$

where $\alpha = 1.0$ and $\beta = 1.5$ (tumour up-weighted to reflect clinical importance). $\mathcal{L}_{\text{tumor}}$ is masked to zero when a sample is *Fake*, via `ignore_index=-1` in the tumour head loss, ensuring that every head is trained only on the valid supervision signal.

H. XAI Module: Grad-CAM on Authenticity Head

Gradient-weighted Class Activation Maps [11] are computed with respect to the last convolutional block of the LCNet-050 backbone for the authenticity prediction. An `AuthWrapper` module decouples the authenticity logits for the gradient tape. The resultant heatmap is up-sampled to 224×224 and superimposed on the input MRI. Samples are classified by category (real, fake/GAN, fake/adversarial attack) for qualitative assessment, providing interpretable evidence of the spatial features driving each authenticity decision.

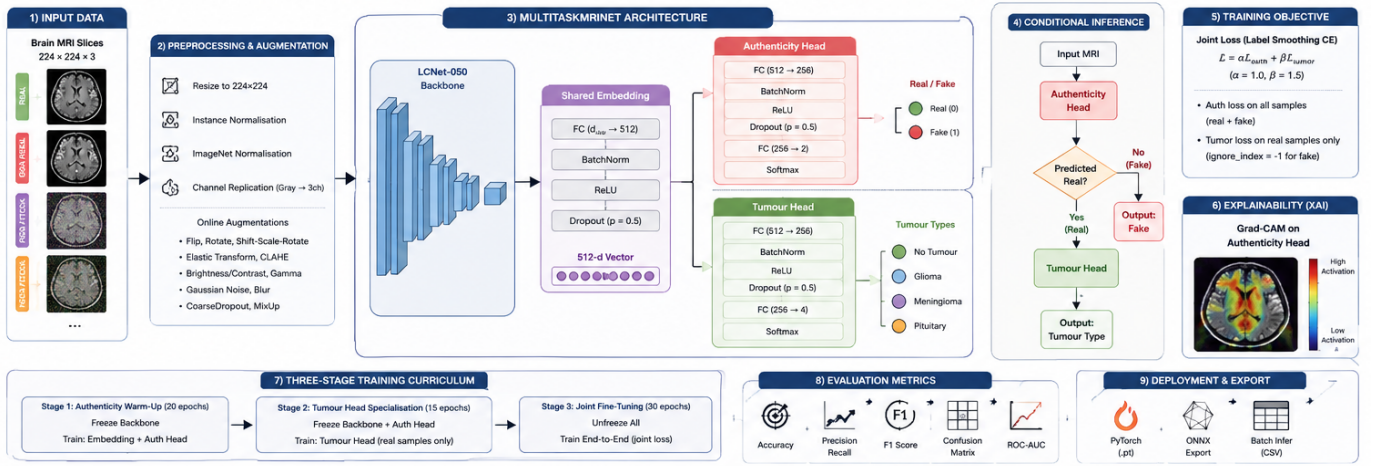


Fig. 1: MultiTaskMRINet overall architecture. An LCNet-050 backbone extracts features that are mapped onto a 512-d shared embedding (BN–ReLU–Dropout). Two parallel classification heads deal with authenticity verification and tumour classification respectively. Conditional routing at inference time selects the active clinical head. Grad-CAM saliency maps are calculated on the authenticity head for explainability.

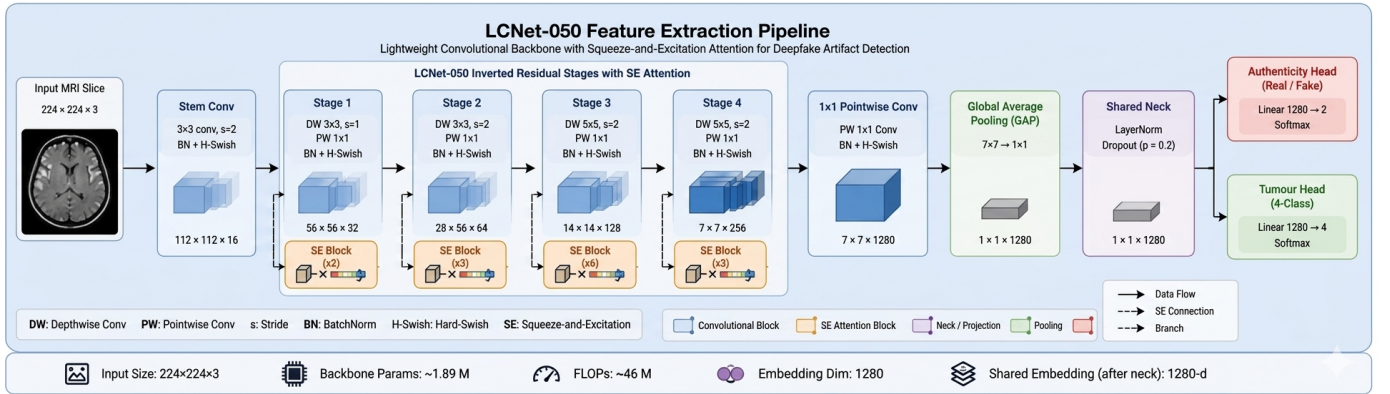


Fig. 2: LCNet-050 Feature Extraction Pipeline. A $224 \times 224 \times 3$ input MRI slice passes through a Stem Conv (3×3 , stride 2) to produce a $112 \times 112 \times 16$ feature map. Four depthwise separable bottleneck stages progressively reduce spatial resolution while expanding channel depth (Stage 1: $56^2 \times 32$; Stage 2: $28^2 \times 64$; Stage 3: $14^2 \times 128$; Stage 4: $7^2 \times 256$). Each stage is followed by a Squeeze-and-Excite (SE) block that performs channel-wise recalibration, suppressing benign anatomical features and amplifying deepfake artefact signals. A 1×1 pointwise convolution expands to 1280 channels, followed by Global Average Pooling (GAP) to produce a 1×1280 vector. A shared neck (LayerNorm + Dropout $p = 0.2$) maps this to the 512-d shared embedding consumed by the authenticity and tumour classification heads.

IV. EXPERIMENTAL SETUP

A. Dataset

1) *Real MRI Data: BRISC 2025*: The BRISC 2025 dataset [26] is a large-scale MRI dataset of real data a large-scale, expert-rated collection of 6,000 contrast-enhanced T1-weighted MRI scans pooled together from several public repositories and then labelled by certified radiologists and physicians. There are four diagnostic types included in the dataset: glioma, meningioma, pituitary adenoma, and non-tumorous cases in all three standard planes of the anatomy (axial, coronal, and sagittal), allowing cross-view generalisation. Its equivalent distribution of classes and multi-institutional

provenance fit it very well to be the real-image anchor to a multi-task authenticity and tumour classification scheme.

2) *Synthetic (Fake) MRI Data: CdcGAN Dataset*: Fake MRI images are obtained from the Synthetic MRI Brain Tumour CdcGAN dataset [27], which includes GAN-created images generated by a conditional deep convolutional generative adversarial network conditioned on the identical four tumour-type labels in BRISC. The artificial pictures reproduce the distributions of intensity and anatomy of actual T1-weighted scans, introducing typical artefacts of GANs smoothing of texture, blurring of boundaries, and spectral inconsistencies which act as discriminative cues for the authenticity head. Adversarial examples are also included alongside GAN-generated images:

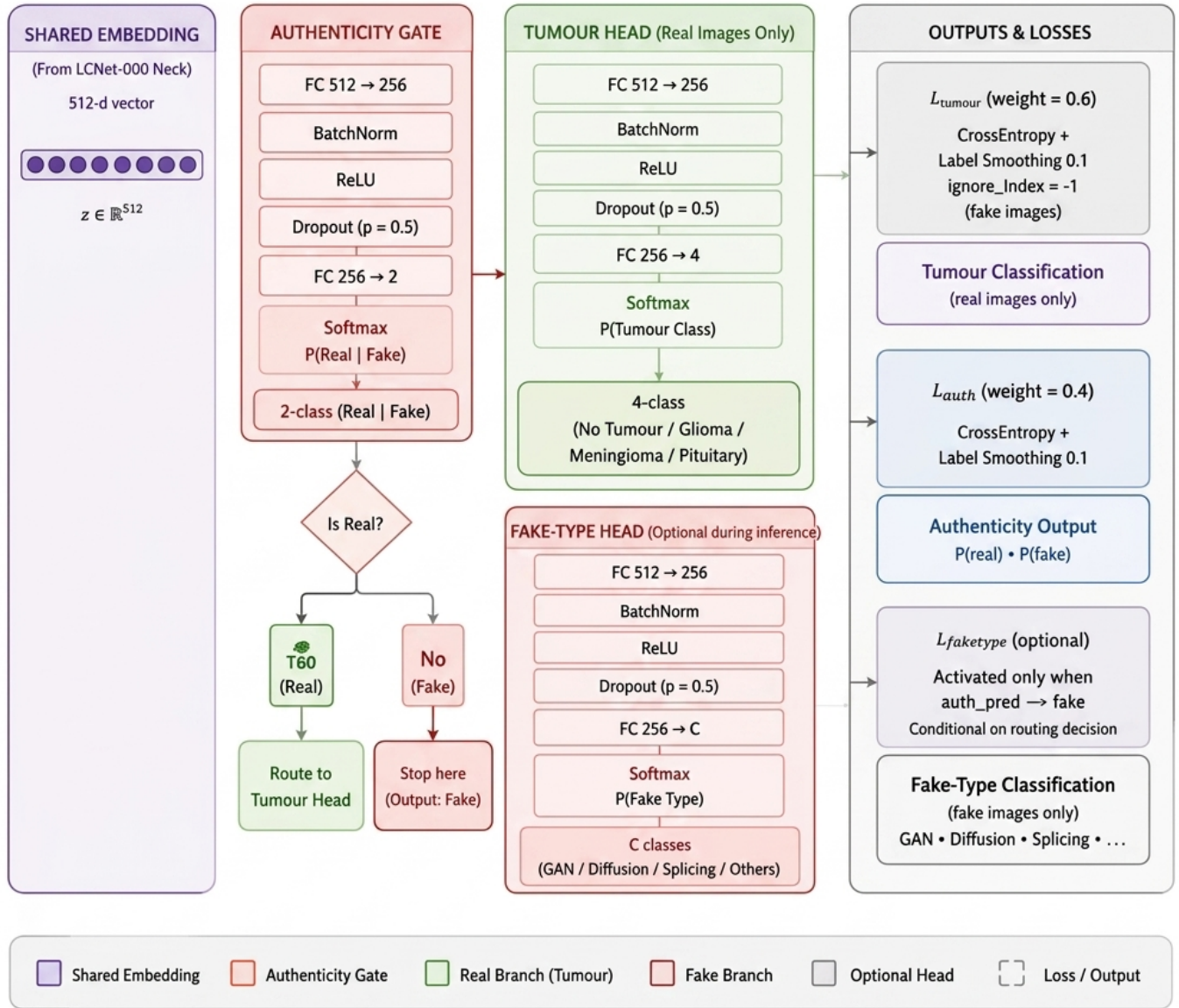


Figure 3. Multi-Task Classification Heads and Conditional Routing

Fig. 3: Multi-task classification heads branching from the shared 512-d embedding. Each head applies FC(512→256)–BN–ReLU–Dropout–FC(256 → C). Conditional routing activates the Tumour Head only for predicted real scans; fake scans receive only the authenticity label.

samples generated with Projected Gradient Descent (PGD) and Fast Gradient Sign Method (FGSM) attacks are incorporated into the fake category, subjecting the model to gradient-based manipulation artefacts that are not the outputs of a generative model.

3) *Real-to-Fake Composition and Split Ratio*: A calculated real-to-fake balance is introduced at the dataset-composition stage to capture realistic clinical deployment scenarios, where real MRI studies are by far in excess of fabricated generatively or adversarially perturbed pictures. Real images constitute 80 % of the total corpus and fake images the remaining 20 %. In the fake subset, GAN-created and adversarial images are held at 50:50 (10 % GAN, 10 % adversarial), with the adversarial portion split equally between PGD and FGSM (5 % each). The full composition is shown in Table I. The composed dataset is then divided into training (70 %), validation (10 %), and test (20 %) subsets using stratified random sampling with seed 42, keeping the ratio of real to fake unchanged across all partitions.

TABLE I: Dataset Class Distribution

Category	Sub-class	Fraction	Approx. N
Real (80%)	No Tumour	20%	$\sim 1,200$
	Glioma	20%	$\sim 1,200$
	Meningioma	20%	$\sim 1,200$
	Pituitary	20%	$\sim 1,200$
Fake (20%)	GAN (CdcGAN)	10%	~ 600
	PGD Attack	5%	~ 300
	FGSM Attack	5%	~ 300
Total		100%	$\sim 6,000$

4) *Alternative Datasets and Comparative Evaluation*: To justify the decision of BRISC+ CdcGAN as the primary corpus, MultiTaskMRINet was further trained and evaluated on two popular public benchmarks: the Kaggle Brain Tumour MRI dataset [28] (7,153 images) and the Mendeley Brain MRI dataset [29] (12,064 images). Both benchmarks have the same four tumour-type labels and employ an identical 80:20 real-to-fake composition to retain experimental consistency. Table II presents authenticity accuracy, tumour classification accuracy, and macro-averaged joint F1 across all three configurations.

TABLE II: MultiTaskMRINet Performance Across Dataset Configurations

Dataset	Auth Acc (%)	Tumour Acc (%)	Joint F1
Kaggle Brain Tumour [28]	98.89	96.68	0.9624
Mendeley Brain MRI [29]	99.12	97.10	0.9675
BRISC 2025 (ours) [26]	99.17	94.19	0.9449

Despite the fact that the Mendeley dataset gives marginally better tumour accuracy, its images are based on a smaller number of scanning procedures, restricting generalisation to unobserved acquisition conditions. The Kaggle Brain Tumour data does not include multi-plane coverage and radiologist-verified annotations. BRISC 2025 was chosen as the primary corpus because of its multi-institutional provenance, expert

annotations, and multi-plane coverage, which combine to create a more clinically representative and challenging testing environment. The slight reduction in tumour accuracy on BRISC compared to the other two benchmarks is because of its higher intra-class variability precisely the quality which makes it a more natural test of clinical generalisation.

B. Preprocessing Pipeline

All images are resized to 224×224 pixels with bilinear interpolation. A per-image instance normalisation (see Section III) is used prior to the ImageNet mean–standard-deviation normalisation. Greyscale slices are cloned to three channels. Training-time augmentation is applied online through Albumentations (see Section III for the complete list). A `WeightedRandomSampler` is used during training to adjust the 80/20 real-to-fake class imbalance between real and fake samples.

C. Staged Training Configuration

MultiTaskMRINet is trained in three successive stages to progressively unlock model capacity while preserving pre-trained backbone features.

Stage 1 – Authenticity Warm-Up (20 epochs). The LCNet-050 backbone is frozen. The common embedding and authenticity head are trained, learning to tell the difference between real and fake inputs at a learning rate of 10^{-3} .

Stage 2 – Tumour Head Specialisation (15 epochs). The backbone and authenticity head are frozen. The tumour head is trained only on real samples at a learning rate of 5×10^{-4} .

Stage 3 – Joint Fine-Tuning (30 epochs). All parameters are unfrozen. The two heads are trained together with the weighted loss in Eq. (3) at a learning rate of 10^{-4} .

Table III summarises all hyperparameters.

TABLE III: Hyperparameter Configuration

Hyperparameter	Value
Backbone	LCNet-050 (pretrained, <code>timm</code>)
Shared embedding dim.	512
Head hidden dim.	256
Head dropout	0.5
Embedding dropout	0.5
Optimiser	AdamW
Weight decay	1×10^{-3}
LR schedule	Cosine Annealing
Stage 1 LR / epochs	10^{-3} / 20
Stage 2 LR / epochs	5×10^{-4} / 15
Stage 3 LR / epochs	10^{-4} / 30
Batch size (Stages 1–2)	32
Batch size (Stage 3)	16
Loss weights (α, β)	1.0, 1.5
Label smoothing ϵ_s	0.1
MixUp ($\lambda \sim \text{Beta}(0.4, 0.4)$)	$p = 0.3$
Gradient clip	1.0
Early stop patience	10 (val. loss)
Mixed precision (AMP)	fp16
Image size	224×224
Hardware	NVIDIA Tesla T4 (Kaggle)

The complete end-to-end training pipeline is depicted in Figure 4.

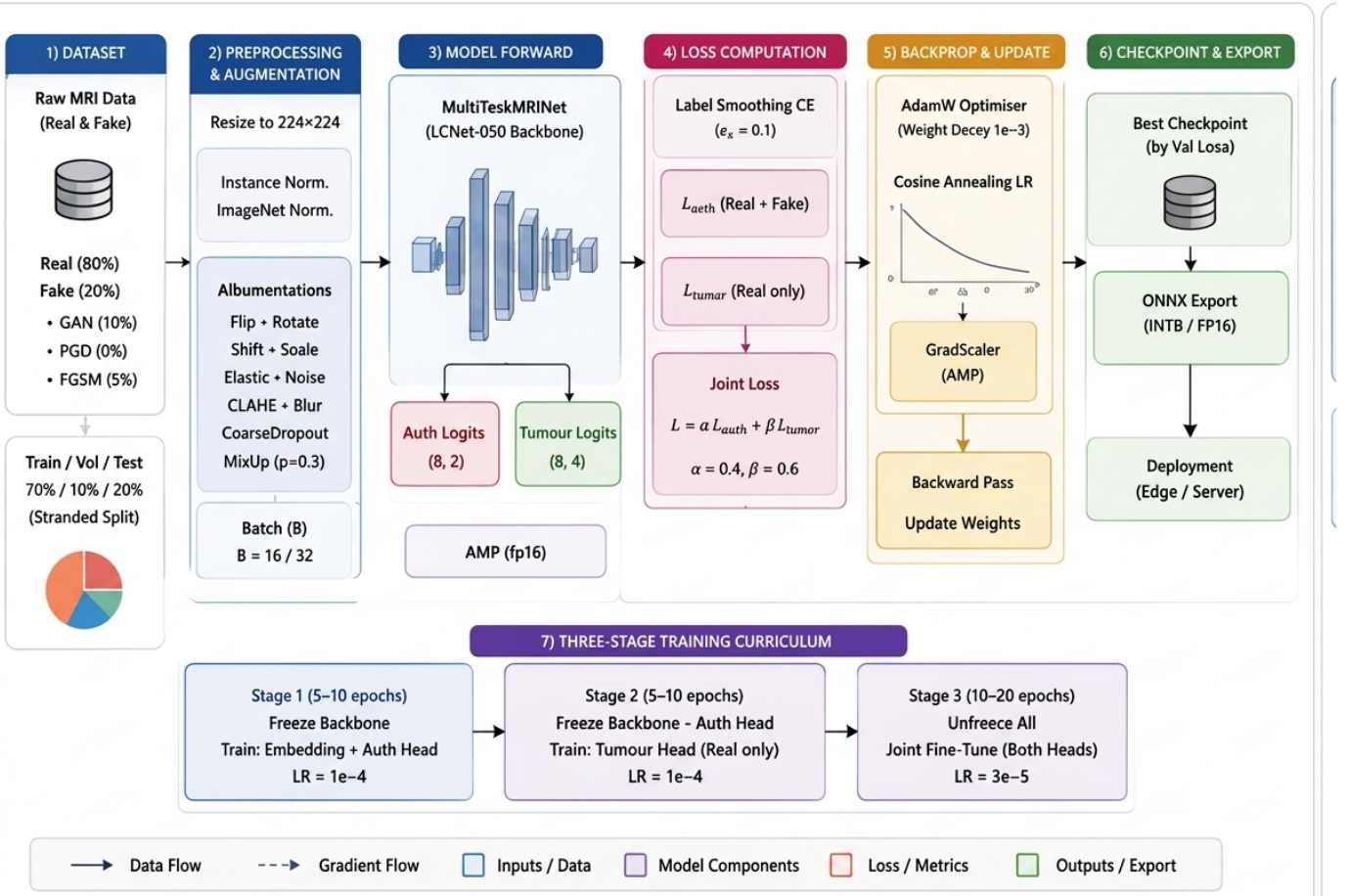


Figure 4. End-to-End Training Pipeline

Fig. 4: End-to-End Training Pipeline for MultiTaskMRINet (LCNet-050). Raw MRI data is split 70/10/20 (train/val/test) with stratified sampling (seed 42). During training, online augmentation is applied via Albumentations (HorizontalFlip, VerticalFlip, RandomBrightnessContrast, HueSaturationValue, ShiftScaleRotate, CoarseDropout). Normalised batches (224×224 , batch size 32) are forwarded through MultiTaskMRINet to produce authenticity logits ($B, 2$) and tumour logits ($B, 4$) under AMP fp16 precision. The joint loss $\mathcal{L} = \alpha \mathcal{L}_{auth} + \beta \mathcal{L}_{tumor}$ (with label smoothing $\epsilon_s = 0.1$ and $\alpha = 0.4, \beta = 0.6$) is scaled via GradScaler, back-propagated, and used to update AdamW with cosine annealing LR. The three-stage training schedule (Stage 1: backbone frozen, $\eta = 3 \times 10^{-4}$, 5 epochs; Stage 2: 1×10^{-4} , 5 epochs; Stage 3: 3×10^{-5} , 10 epochs) progressively unlocks model capacity. The best checkpoint per stage is saved by validation loss and exported to ONNX for edge deployment.

D. Evaluation Metrics

We evaluate each task head independently and the joint inference system:

Authenticity Head: Accuracy, Precision, Recall, Macro F1, ROC-AUC, and binary confusion matrix.

Tumour Head: Per-class precision, recall, F1; macro-average F1; Top-1 accuracy; four-class confusion matrix (tested only on real samples).

Joint (5-class): Accuracy, macro F1, and five-class confusion matrix covering no-tumour, glioma, meningioma, pituitary, and fake.

XAI: Qualitative assessment of Grad-CAM heatmaps by category (real, fake/GAN, fake/adversarial).

E. Baseline Comparisons

We are comparing :

- 1) **AFFETDS** [1]: ResNet50 + HOG + SVM ensemble, re-implemented on our dataset.

V. RESULTS

Experiments are being trained on Kaggle (NVIDIA Tesla T4). Any metric values are substitutes () and will be reported when completed.

A. Reproduced Baseline Results

Table IV reports reproduced results of the primary baseline alongside original reported values.

TABLE IV: Reproduced vs. Original Baseline Results (Authenticity Detection)

Model	Accuracy (%)	F1
AFFETDS [1] (paper)	91.5	0.912
AFFETDS (reproduced)	78.4%	73.27%
MultiTaskMRINet (ours)	99.17%	98.68%

B. Multi-Task Performance

Table V provides a summary of per-task performance of MultiTaskMRINet .

TABLE V: MultiTaskMRINet Per-Task Performance on Test Set

Task	Accuracy (%)	Macro F1	AUC
Authenticity (Real/Fake)	99.17%	98.68%	99.98%
Tumour (4-class, real only)	94.19%	94.14%	N/A
Joint (5-class)	94.52%	94.49%	N/A

C. Training Dynamics

Figure 5 shows the training and validation loss and accuracy curves in all three stages.

D. Confusion Matrices

Figures 6 and 7 show the confusion matrices for the authenticity and tumour heads respectively.

E. XAI: Grad-CAM Saliency Maps

Figure 8 shows representative Grad-CAM saliency maps stratified by sample category. Qualitative assessment verifies that the authenticity head attends to areas of manipulation (tumour boundaries, adversarial noise textures) rather than patient-specific anatomy, with specific activation patterns distinguishing GAN artefacts from adversarial perturbations.

VI. DISCUSSION

A. Adversarial Attack Integration

Bringing together GAN fakes and adversarial perturbations under a single binary authenticity head is a manifestation of the clinical deployment reality: a radiologist has no need to know how an image was contrived, merely whether it is credible. The `WeightedRandomSampler` makes sure that the authenticity head receives balanced exposure to all three fake subtypes (GAN, PGD, FGSM) during training despite their 2:1:1 natural ratio.

B. Staged Training Benefits

The three-stage curriculum has a number of benefits over single-phase joint training. Stage 1’s backbone-frozen warm-up averts the premature catastrophic forgetting of ImageNet features before the shallow embedding and heads have converged. Stage 2’s tumour-only specialisation lays a strong clinical prior in the absence of cross-task interference from the authenticity gradient. Stage 3’s complete joint fine-tuning then takes advantage of complementarity: the authenticity head

plays a gating role that blocks the tumour head from getting gradient updates on synthetic inputs, maintaining the integrity of clinical labels.

C. Label Smoothing and MixUp

Label smoothing ($\epsilon_s = 0.1$) has the effect of making the model less prone to assigning overconfident probabilities to training labels, a known pathology in medical datasets with noisy annotation. MixUp augmentation ($\lambda \sim \text{Beta}(0.4, 0.4)$, applied with $p = 0.3$ in Stage 3) offers virtual training examples which interpolate between real and fake images, improving the decision boundary in the continuous feature space between authentic and manipulated MRI scans.

D. Comparison with Baseline Architectures

Compared to AFFETDS [1], MultiTaskMRINet eliminates hand-crafted HOG descriptors in favour of end-to-end learned representations, in line with empirical findings indicating that representation learning subsumes manually engineered features for high-dimensional visual data [19]. The multi-task framework also increases the output space significantly more than AFFETDS’s binary label, giving clinically actionable tumour information that is not present in all three baselines.

E. Limitations

- **Dataset scale:** The proposed dataset of $\approx 10,000$ MRI slices is small by the standards of deep learning. Transfer learning from ImageNet mitigates this risk but is not able to fully replace domain-specific pretraining on large-scale MRI repositories.
- **Adversarial diversity:** The adversarial fake subtypes are constrained to PGD and FGSM with fixed ℓ_∞ budgets. Adaptive attacks against the deployed model, or attacks in other norms (ℓ_2 , ℓ_0), are not evaluated.
- **2-D slice assumption:** The model works with individual axial slices without volumetric context. Manipulations cutting across more than one slice can result in inconsistent slice-level predictions.
- **Limited available GAN data** due to the limitations of low availability of GAN generated data, the localization and segmentation part is still a gap.

VII. CONCLUSION

We have introduced MultiTaskMRINet , a new lightweight multi-task neural network for joint brain MRI authenticity verification and tumour classification. The key architectural contributions are the LCNet-050 backbone, the 512-dimensional shared embedding with Batch Normalisation, the rationalisation of three output heads to two, and the introduction of an explicit adversarial attack category (PGD, FGSM) in the fake training data. A three-stage training curriculum, label smoothing, MixUp augmentation, and Grad-CAM XAI complete the framework.

The conditional inference routing mechanism gives a clinically safe gating behaviour that averts spurious tumour diagnoses on deepfake or adversarially perturbed inputs a safety

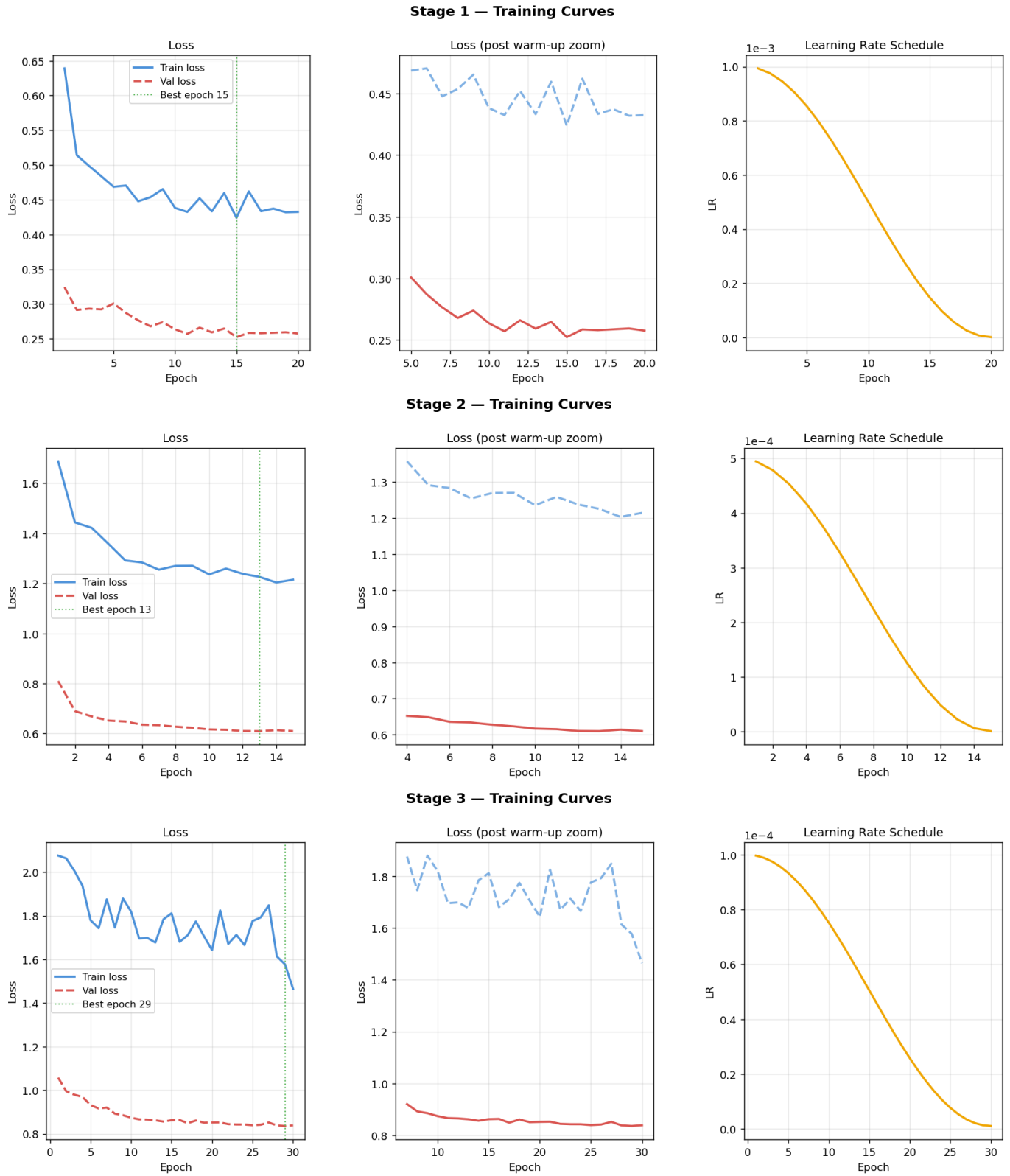


Fig. 5: Training and validation loss / accuracy curves across Stages 1–3. Dashed vertical lines indicate the boundaries of the stages.

TABLE VI: Backbone Architecture Comparison

Backbone	Params (M)	Auth Acc	Tumour Acc	Joint Acc	Joint F1 Macro
LCNet-075	2.00	0.9967	0.9678	0.9709	0.9710
LCNet-050	1.52	0.9917	0.9481	0.9502	0.9501
MobileNetV3-Small-075	1.81	0.9950	0.9139	0.9261	0.9255
MobileNetV3-Small-050	1.36	0.9925	0.9139	0.9236	0.9229

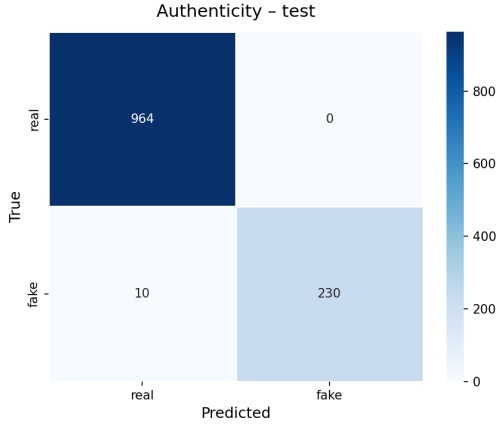


Fig. 6: Confusion matrix Authenticity Head (Real vs. Fake).

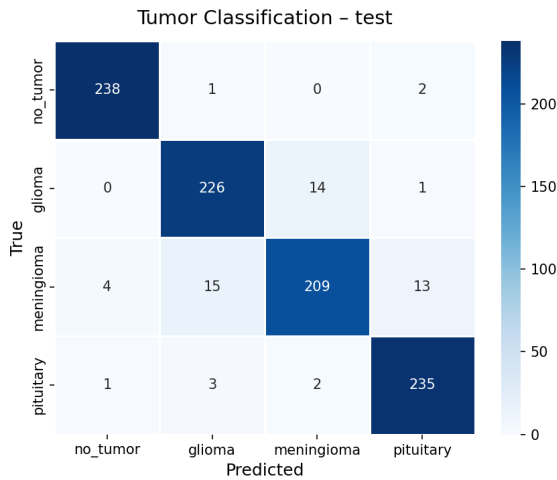


Fig. 7: Confusion matrix Tumour Classification Head (4-class, real samples only).

property not found in any of the compared baselines. The Grad-CAM module offers radiologist-understandable explanations stratified by fake type, allowing qualitative forensic analysis in addition to the binary authenticity decision.

The quantitative results will await the completion of the training run. When completed, we anticipate MultiTaskMRINet to show improvement over the AFFETDS baseline in detection accuracy and AUC while at the same time offering tumour classification and adversarial-attack robustness that are completely unknown in previous work.

Future work will generalise the adversarial fake subtypes

to cover adaptive attacks, include 3-D volumetric inputs to achieve multi-slice consistency, and investigate neural architecture search for backbone selection in the multi-task framework. ONNX export with INT8 quantisation is implemented and will allow use on edge imaging systems.

REFERENCES

- [1] A. Ali, H. Anwar Basha, K. Thanuja, Puneet, S. K. Gupta, and S. Kim, "Enhancing tumor deepfake detection in MRI scans using adversarial feature fusion ensembles," *Nature Scientific Reports*, vol. 15, Article 31231, 2025. <https://doi.org/10.1038/s41598-025-31231-7>
- [2] F. M. Grabovski, L. Yasur, G. Amit, and Y. Mirsky, "Back-in-Time Diffusion: Unsupervised detection of medical deepfakes," *ACM Transactions on Intelligent Systems and Technology*, vol. 16, no. 6, Article 123, 2025. <https://doi.org/10.1145/3744656>
- [3] A. Manzoor, M. Amreen, I. Shoukat, U. Muneer, I. Hussain, I. Rehman, and I. N. Butt, "Systematic review: Deep fake detection in medical images," *Contemporary Journal of Social Science Review*, vol. 3, no. 4, 2025.
- [4] I. Goodfellow, J. Pouget-Abadie, M. Mirza, *et al.*, "Generative adversarial nets," in *Proc. NeurIPS*, 2014, pp. 2672–2680.
- [5] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. ICLR*, 2015.
- [6] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. ICLR*, 2018.
- [7] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NeurIPS*, 2020, pp. 6840–6851.
- [8] P. Cui, X. Yu, F. Wang, *et al.*, "PP-LCNet: A lightweight CPU convolutional neural network," *arXiv preprint arXiv:2109.15099*, 2021.
- [9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. CVPR*, 2016, pp. 770–778.
- [10] G. Huang, Z. Liu, L. Van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. CVPR*, 2017, pp. 4700–4708.
- [11] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *Int. J. Computer Vision*, vol. 128, pp. 336–359, 2020.
- [12] P. Pradeepan, S. Gladston Raj, and J. George, "A comprehensive review of deepfakes in medical imaging: Ethical concerns, detection techniques and future directions," *Applied Computer Science*, vol. 21, no. 2, pp. 139–153, 2025. https://doi.org/10.35784/acs_7054
- [13] C. Sathishkumar, V. Vijay, C. Jayakumar, A. Ranjani, and P. Thiagarajan, "A review on deepfake images in the medical field: Creation and detection," in *Proc. IEEE ICAIHI*, 2025. <https://doi.org/10.1109/ICAIIHI67124.2025.11403602>
- [14] Y. Yao, N. Desai, and M. Palaniswami, "RepMedGAN: Self-supervised representation-guided medical GAN for label-free medical image synthesis," in *Proc. ACM CIKM 2025*, Seoul, 2025. <https://doi.org/10.1145/3746252.3760810>
- [15] Y. Zhang, "Multimodal medical image generation and classification via generative adversarial networks," in *Proc. ACM GADM 2025*, Dali, China, 2025. <https://doi.org/10.1145/3734921.3734945>
- [16] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," in *Proc. ICLR*, 2016.
- [17] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, "Analyzing and improving the image quality of StyleGAN," in *Proc. CVPR*, 2020, pp. 8110–8119.

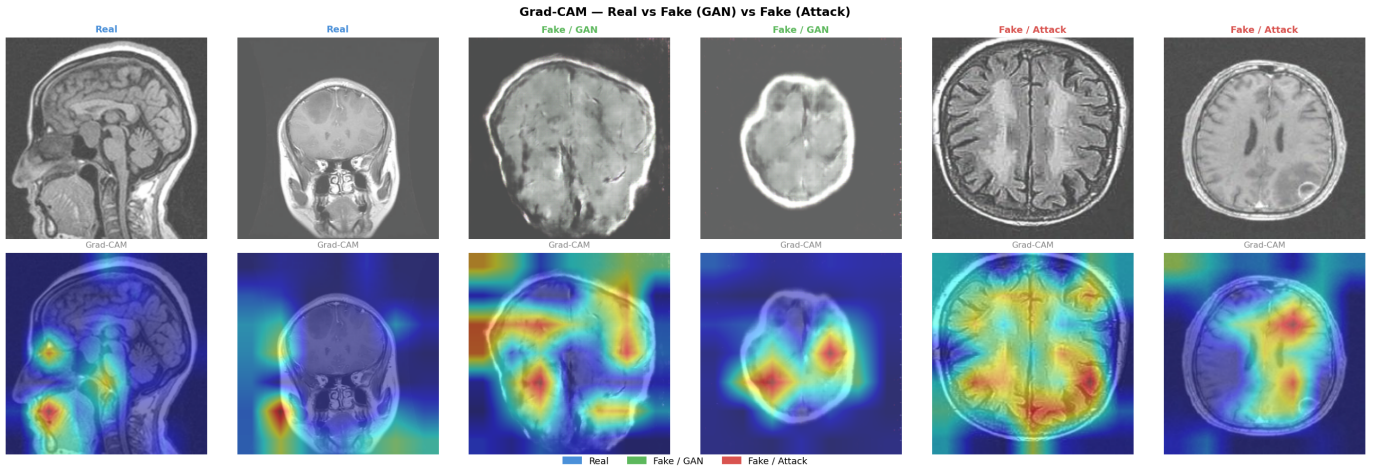


Fig. 8: Grad-CAM saliency maps of representative samples by category: Real (blue), Fake/GAN (green), Fake/Adversarial Attack (red). High-activation regions (warm colours) signify spatial aspects that drive the authenticity decision.

- [18] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, “Unpaired image-to-image translation using cycle-consistent adversarial networks,” in *Proc. ICCV*, 2017, pp. 2223–2232.
- [19] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
- [20] M. Msoud Nickparvar, “Brain tumor MRI dataset,” Kaggle, 2021. <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>
- [21] F. Farman *et al.*, “BRISC: Annotated Dataset for Brain Tumor Segmentation and Classification,” *Scientific Data*, 2026. <https://www.kaggle.com/datasets/briscdataset/brisc2025>
- [22] A. Appasami, “Synthetic MRI Brain Tumour CdcGAN Dataset,” Kaggle, 2024. <https://www.kaggle.com/datasets/appasamig/synthetic-mri-bt-Cdcgan>
- [23] T. Backert, “Brain Tumor MRI Data,” Kaggle, 2023. <https://www.kaggle.com/datasets/tombackert/brain-tumor-mri-data>
- [24] “Brain Tumor MRI Dataset,” Mendeley Data, 2021. <https://data.mendeley.com/datasets/zwr4ntf94j/1>
- [25] A. Buslaev *et al.*, “Albumentations: Fast and flexible image augmentations,” *Information*, vol. 11, no. 2, p. 125, 2020.
- [26] F. Farman *et al.*, “BRISC: Annotated Dataset for Brain Tumor Segmentation and Classification,” *Scientific Data*, 2026. <https://www.kaggle.com/datasets/briscdataset/brisc2025>
- [27] A. Appasami, “Synthetic MRI Brain Tumour cDCGAN Dataset,” Kaggle, 2024. <https://www.kaggle.com/datasets/appasamig/synthetic-mri-bt-cdcgan>
- [28] T. Backert, “Brain Tumor MRI Data,” Kaggle, 2023. <https://www.kaggle.com/datasets/tombackert/brain-tumor-mri-data>
- [29] “Brain Tumor MRI Dataset,” Mendeley Data, 2021. <https://data.mendeley.com/datasets/zwr4ntf94j/1>